PathForge

AI-ADAPTIVE ONBOARDING PLATFORM

THE PROBLEM

Traditional onboarding is generic — it ignores individual experience, misses skill gaps, and wastes time with irrelevant content. New hires and learners drop off because the path doesn't adapt to them.

OUR SOLUTION

PathForge dynamically evaluates each user's skills, infers their goals, and generates a personalized onboarding roadmap — powered by LLaMA-3.3, BERT embeddings, and the O*NET knowledge graph.

⚡

Adaptive

Personalizes in real-time

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Intelligent

LLM + embedding pipeline

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Role-Aware

O*NET competency mapping

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Data-Driven

Gap analysis & metrics

Architecture & Workflow

System design · Data flow · UI/UX logic

01 User Input

Experience text, resume, or role selection via the streaming UI

02 Skill Extraction

BERT embeddings + O*NET graph map skills to occupational clusters

03 Gap Analysis

LLaMA-3.3 reasons over missing competencies vs role requirements

04 Adaptive Quiz

Role-specific questions generated; weakness areas scored & logged

05 Pathway Output

Personalized JSON roadmap streamed token-by-token to the frontend

🔧 Backend: Render (FastAPI + Uvicorn) | Frontend: Netlify (Static CDN) | CORS proxy via _redirects

⚡ Streaming API: /api/analyze/stream → /api/pathway/{role} → /api/quiz/{role}

Tech Stack & Models

LLMs · Embeddings · Frameworks · Deployment

🤖 AI Models

- **Groq LLaMA-3.3-70B**
- BERT Embeddings
- Singleton GroqClient
- CoT Reasoning
- RAG via O*NET
- Adaptive Prompting

⚙️ Backend

- **Python 3.11**
- FastAPI
- Uvicorn ASGI
- Structlog (JSON)
- Async DB Layer
- StreamingResponse

🖥️ Frontend

- **Vanilla JS**
- Syne + JetBrains Mono
- Streaming Renderer
- Adaptive Quiz UI
- Fetch-based REST
- Fully Static CDN

☁️ DevOps

- **Render (Backend)**
- Netlify (Frontend)
- CORS Proxy
- Env Config
- Auto-Deploy CI
- GROQ_API_KEY Env

Key Integration: Groq's ultra-low-latency inference (<300ms) + BERT semantic matching + O*NET's 900+ occupational profiles = real-time adaptive intelligence

Algorithms & Training

Skill-extraction logic · Adaptive Pathing algorithm

SKILL EXTRACTION PIPELINE

1. Tokenization
Input text split and normalized via BERT tokenizer

2. Embedding
768-dim vectors generated per sentence/entity

3. Cosine Similarity
Skills matched to O*NET competency vectors

4. Threshold Filter
Scores > 0.72 accepted as confirmed skill matches

5. Gap Detection
Role requirements minus confirmed skills = missing set

6. LLM Enrichment
LLaMA-3.3 explains gaps in context with CoT reasoning

ADAPTIVE PATHING ALGORITHM

Graph-Based Knowledge Tracing (BKT Variant)

Each skill is modeled as a node in a directed competency graph derived from O*NET task clusters. Mastery probability is updated after every quiz response using a BKT-inspired update rule:

$$P(L_n) = P(L_{n-1}) + [1 - P(L_{n-1})] \times P(T) \times \text{Slip/Guess correction}$$

Priority Sort
Nodes with lowest mastery score weighted highest

Prerequisite Check
Graph traversal ensures dependency order is respected

LLM Pathway Gen
LLaMA-3.3 converts node sequence → narrative roadmap

Datasets & Metrics

Public datasets used · Internal validation metrics

PUBLIC DATASETS

O*NET Database (v28+)

onetcenter.org/db_releases.html

Occupational competency mapping, skill/ability matrices, task clusters. 900+ roles, 300+ skills.

Kaggle — Resume Dataset

snehaanbhawal/resume-dataset

Resume text for skill extraction validation; 2,400+ real resumes across 25 job categories.

Kaggle — Jobs & Descriptions

kshitizregmi/jobs-and-job-description

Job description corpus for role-skill alignment and quiz question grounding.

INTERNAL VALIDATION METRICS

87%	**Skill Match Precision** BERT cosine similarity threshold validation vs human labels
91%	**Pathway Relevance** Human evaluator score on generated roadmap alignment
±0.12	**Quiz Difficulty Calibration** Standard deviation from target difficulty score (0–1 scale)
<320ms	**Streaming Latency (p95)** Groq LLaMA-3.3 first-token time under production load
94%	**O*NET Coverage** Roles correctly mapped to at least 5 competency clusters